

**Plasmon-mediated pairing in 2D:
the BCS gap equation lives at small angles, and what remains once the
small-angle sector is removed**

0. Setup and the one number to keep in mind

Notation: $\alpha = e^2/\epsilon_\infty v_F$; N flavours; screening momentum $\kappa = N\alpha k_F$; 2D plasmon $\Omega_q^2 = \frac{1}{2}\kappa v_F^2 q$; plasmon-window cutoff $K = a\kappa$, $a \lesssim 1$. Plasmon-pole interaction, valid for $|\nu| \gg v_F q$:

$$U(i\nu, q) = \frac{2\pi e^2}{\epsilon_\infty q} \frac{\nu^2}{\nu^2 + \Omega_q^2}. \quad (1)$$

Linearized gap equation at $T = 0$, $\mathbf{k}$ on the Fermi surface, external frequency w :

$$\Delta(w) = - \int \frac{dw'}{2\pi} \int \frac{d^2 p}{(2\pi)^2} U(w - w', |\mathbf{k} - \mathbf{p}|) \frac{\Delta(w')}{w'^2 + \xi_p^2}. \quad (2)$$

Units $v_F = \kappa = 1$ (frequencies in κv_F , momenta in κ), so $e^2/\epsilon_\infty = \alpha$, $\Omega_q^2 = q/2$, $K = a$, $\Omega_K^2 = a/2$, and—this is the number to keep in mind—

$$k_F = \frac{\kappa}{N\alpha} = \frac{1}{N\alpha} \gg 1 \quad (\text{weak coupling}). \quad (3)$$

A momentum transfer q between two states on the Fermi surface is a scattering angle $\phi = q/k_F = N\alpha q$ (in units of κ); if one of the states is off the Fermi surface, $\phi < q/k_F$ (Section 1.1). Every momentum transfer that appears below is $q \leq K = a$, hence every scattering angle is

$$\phi \leq \phi_K \equiv \frac{K}{k_F} = aN\alpha \ll 1. \quad (4)$$

Sections 1–2 make this explicit and explain why it is a problem; Section 3 removes the small-angle sector and computes the critical coupling that results.

1. The naive (BCS-like) gap equation and where it lives

1.1 Momentum integral, done exactly. Write $p = k_F + \xi/v_F$ and let ϕ be the angle between $\mathbf{p}$ and $\mathbf{k}$ ($|\mathbf{k}| = k_F$). For $\phi \ll 1$ and $|\xi| \ll v_F k_F$,

$$q^2 = |\mathbf{k} - \mathbf{p}|^2 \simeq \xi^2 + k_F^2 \phi^2. \quad (5)$$

The measure is $d^2 p = p dp d\phi \simeq k_F d\xi d\phi$ (units $v_F = 1$). Change variables $\phi \rightarrow q$ at fixed ξ : $k_F^2 \phi d\phi = q dq$, i.e. $d\phi = q dq / k_F \sqrt{q^2 - \xi^2}$, with two branches $\phi \gtrless 0$. Therefore

$$\begin{aligned} \int \frac{d^2 p}{(2\pi)^2} \frac{U(\nu, q)}{w'^2 + \xi^2} &= \frac{k_F}{(2\pi)^2} \int d\xi \int d\phi \frac{U}{w'^2 + \xi^2} = \frac{2}{(2\pi)^2} \int d\xi \int_{|\xi|}^K \frac{dq q U(\nu, q)}{\sqrt{q^2 - \xi^2} (w'^2 + \xi^2)} \\ &= \frac{1}{2\pi^2} \int_0^K dq q U(\nu, q) \int_{-q}^q \frac{d\xi}{(w'^2 + \xi^2) \sqrt{q^2 - \xi^2}} = \frac{1}{2\pi} \int_0^K dq \frac{q U(\nu, q)}{|w'| \sqrt{w'^2 + q^2}}. \end{aligned} \quad (6)$$

In the second line the order of integration was exchanged ($|\xi| < q < K$ is the same region as $0 < q < K$, $|\xi| < q$), and the ξ -integral was done with $\xi = q \sin \theta$: $\int_{-\pi/2}^{\pi/2} d\theta / (w'^2 + q^2 \sin^2 \theta) =$

$\pi/|w'|\sqrt{w'^2 + q^2}$. Note that k_F cancelled: (6) does not know about k_F , which is another way of saying that it is a small-angle expression.

Caution: after the ξ -integration q is no longer an angle. The variable change $\phi \rightarrow q$ was made at fixed ξ , and by (5) a given q corresponds to

$$|\phi| = \frac{\sqrt{q^2 - \xi^2}}{k_F}, \quad 0 \leq |\phi| \leq \frac{q}{k_F},$$

with $|\phi| = q/k_F$ only on shell ($\xi = 0$) and $\phi \rightarrow 0$ at $|\xi| \rightarrow q$ (a transfer purely along the Fermi-surface normal). Once ξ has been integrated out in (6), every q contains all angles between 0 and q/k_F . Two consequences. (i) Upper bounds survive: $q \leq K$ implies $|\phi| \leq K/k_F = \phi_K$ for every term, so the statement “everything is at angles below ϕ_K ” is exact. Conditions on the full transfer q (such as the plasmon condition $|\nu| > v_F q$) can likewise be imposed on the q -integral after the ξ -integration without error. (ii) A *lower* cut on q is *not* a lower cut on the angle: $q > q_f$ keeps the forward region $|\phi| < q_f/k_F$ at large $|\xi|$. An angular cutoff has to be imposed before the ξ -integration, as a q -dependent restriction of the ξ -range; this is done in Section 3.1.

The validity condition of the interaction. Eq. (1) holds only for $|\nu| \gg v_F q$, so it enters the gap equation as $U(i\nu, q) \theta(|\nu| - cv_F q)$, with $c \gtrsim 1$ the constant in “ $\gg$ ” (below $c = 1$ unless stated). The θ -function depends on the full transfer q and on $\nu = w - w'$, neither of which involves ξ or ϕ separately, so it simply lowers the upper limit of the q -integral, $K \rightarrow q_{\max}(\nu) \equiv \min(K, |\nu|/cv_F)$. This is exact (point (i) above; I also checked it by imposing the θ -function literally inside a two-dimensional (ξ, ϕ) integration, agreement to 10^{-4}). With $qU = 2\pi\alpha\nu^2/(\nu^2 + q^2/2)$, Eq. (2) becomes

$$\boxed{\begin{aligned} \Delta(w) &= -\frac{\alpha}{2\pi} \int dw' \int_0^{q_{\max}(\nu)} dq \frac{\nu^2}{\nu^2 + \Omega_q^2} \frac{\Delta(w')}{|w'|\sqrt{w'^2 + q^2}}, \\ \nu &= w - w', \quad q_{\max}(\nu) = \min\left(K, \frac{|\nu|}{cv_F}\right). \end{aligned}} \quad (7)$$

Note which frequencies the condition constrains: the *transfer* ν , not the fermion frequency w' . Pairs with $|w'| \ll v_F q$ (where the pair weight is the Cooper weight $1/|w'|v_F q$) exchanging a plasmon with $|\nu| \simeq |w| \gg v_F q$ are inside the validity domain; the truncation addendum discusses the alternative of cutting w' too. All numerical results in this note include the θ -function.

1.2 The momentum transfers that matter, translated into angles. Take the frequency-dependent part of the interaction, $\frac{\nu^2}{\nu^2 + \Omega_q^2} = 1 - \frac{\Omega_q^2}{\nu^2 + \Omega_q^2}$. At fixed transfer frequency ν the retarded piece $\Omega_q^2/(\nu^2 + \Omega_q^2)$ is $\simeq 1$ for $q \gg q_\nu$ and $\simeq q/2\nu^2 \rightarrow 0$ for $q \ll q_\nu$, where

$$q_\nu \equiv \frac{2\nu^2}{\kappa v_F^2} \quad (\Omega_{q_\nu} = |\nu|) \quad \implies \quad \phi_\nu = \frac{q_\nu}{k_F} = 2N\alpha \left(\frac{\nu}{\kappa v_F} \right)^2. \quad (8)$$

So the plasmon dynamics at frequency ν is carried by transfers between q_ν and $q_{\max}(\nu) = \min(K, |\nu|/cv_F)$; this window is non-empty only for $2\nu^2 < |\nu|/c$, i.e. $|\nu| < \kappa v_F/2c$, which is the statement that the plasmon pole must itself satisfy $\Omega_q > cv_F q$. By the caution in 1.1 these are scattering angles below $q/k_F \leq \phi_K = aN\alpha$ (ϕ_ν is the on-shell angle of the transfer q_ν). For the frequencies that matter, $\nu \sim \Omega_K$, both q_ν and $q_{\max}$ are $\sim K$, so the relevant transfers correspond to on-shell angles $\sim aN\alpha$ and to all smaller angles off shell. Nothing in (7) reaches beyond ϕ_K ; nothing in it is large-angle scattering.

1.3 The same statement from the angular average. The Fermi-surface average that many papers use makes this visible in one line. With both momenta on the Fermi surface, $q = 2k_F \sin(\theta/2) \simeq k_F |\theta|$, the cutoff $q < q_{\max}(\nu)$ restricts $|\theta| < q_{\max}/k_F \leq \phi_K$, and $\int_{-\pi}^{\pi} \frac{d\theta}{2\pi} \rightarrow \frac{1}{\pi k_F} \int_0^{q_{\max}} dq$. With $N_0 = k_F/2\pi v_F$ and $\Omega_{q_{\max}}^2 = q_{\max}^2/2$,

$$\lambda(\nu) \equiv N_0 \langle U \theta (|\nu| - cv_F q) \rangle_{\text{FS}} = \frac{\alpha}{\pi} \int_{q_{\min}}^{q_{\max}(\nu)} \frac{dq}{q} \frac{\nu^2}{\nu^2 + \Omega_q^2} \simeq \frac{\alpha}{\pi} \ln \frac{q_{\max}(\nu)}{q_{\min}} - \frac{\alpha}{\pi} \ln \left(1 + \frac{\Omega_{q_{\max}}^2}{\nu^2} \right) \quad (9)$$

(the exact second term is $\frac{\alpha}{\pi} \ln \frac{\nu^2 + \Omega_{q_{\max}}^2}{\nu^2 + \Omega_{q_{\min}}^2}$; the form above assumes $q_{\min} \ll q_{\nu}$). For $|\nu| > cv_F K$ this is the familiar $\frac{\alpha}{\pi} [\ln(K/q_{\min}) - \ln(1 + \Omega_K^2/\nu^2)]$; at lower ν the upper limit is $|\nu|/cv_F$ and both terms shrink (e.g. $K = 0.3\kappa$, $c = 1$: the dynamical term is 1.79 instead of 2.77 in units of α/π at $\nu = 0.1\kappa v_F$). The integrand is $dq/q = d\phi/\phi$: the coupling is a logarithm over scattering angles, the instantaneous part diverging towards $\phi \rightarrow 0$ and the dynamical part receiving equal weight from every e-fold of ϕ between ϕ_{ν} and $q_{\max}(\nu)/k_F$. Again, all of it lives below $\phi_K = aN\alpha$.

1.4 What the solution of (7) is built from. The pair weight in (7) has two regimes: $1/w'^2$ for $|w'| \gg v_F q$ and the ordinary Cooper weight $1/|w'|v_F q$ for $|w'| \ll v_F q$ (the shell $|\xi| < v_F q$ integrated at low frequency). The kernel is positive at all frequencies, so any solution must change sign (Tolmachev–Anderson). Projecting on two frequency blocks, Δ_{IR} below $w_1 \simeq 0.3\Omega_K$ and Δ_{H} above, gives $(1 + \alpha M_{11})(1 + \alpha M_{22}) = \alpha^2 M_{12} M_{21}$ with $M_{21} = m \ln(w_1/E)$, E the infrared scale, and the solution $\alpha^2 M_{12} m \ln(\Omega_K/E) \simeq 1$: a solution exists for any α , with $T_c \sim \Omega_K \exp(-c^2/\alpha^2)$ (previous note; $c \simeq 2.7$ at $a = 0.3$). Every ingredient of this mechanism—the retarded repulsion in M_{12} , the Cooper logarithm in M_{21} —comes from transfers q inside the window, hence from angles below ϕ_K . Figure 1b quantifies it: removing the angles $|\phi| < q_f/k_F$ from (7) (as an angular cut, with the ξ -window of Section 3.1) and recomputing the pairing eigenvalue μ ($a = 0.3$, $E = 10^{-6}\kappa v_F$), one finds that removing the angles below $0.1\phi_K$ already reduces $|\mu|$ by 35%, below $0.45\phi_K$ by a factor 4.5, and below $0.85\phi_K$ by a factor 30.

The clash with BCS. The BCS/Cooper-ladder treatment of a pairing interaction is justified when the interaction transfers momenta of order k_F : then the ladder is the one-loop RG in the Cooper channel and the logarithm it produces is real. Equation (7) is such a ladder, but for an interaction whose transfers are all $\lesssim aN\alpha k_F$. For such small-angle scattering the ladder is *not* the right resummation—this sector can be solved exactly, and the exact answer is that it does not produce a Cooper logarithm at all. That is the content of the next section.

2. The small-angle sector: bosonization and the pair susceptibility

2.1 Which angles are “small”. Let R_s be the range of the interaction, so that it transfers momenta $q \lesssim 1/R_s$. Linearize the dispersion on a patch of the Fermi surface around its central direction: a state at angle ϕ from the patch centre and radial distance $k_{\parallel}$ from the Fermi surface has energy $v_F k_{\parallel} + k_{\perp}^2/2m$ with $k_{\perp} \simeq k_F \phi$. The tangential (curvature) term is negligible against the radial energy scale probed by the interaction, v_F/R_s , and the patch can be treated as flat (which is what makes the sector exactly solvable), when

$$\frac{k_F^2 \phi^2}{2m} \ll \frac{v_F}{R_s} \quad \Longleftrightarrow \quad \phi \ll \phi_0 \equiv \frac{1}{\sqrt{k_F R_s}}. \quad (10)$$

The small parameter of the bosonization is $1/k_F R_s \ll 1$ (interaction range much longer than the Fermi wavelength), and $\phi_0 = (k_F R_s)^{-1/2}$ is the angle separating the exactly solvable small-angle sector from

the large-angle sector where the Cooper ladder is legitimate. (It is the largest admissible patch half-width; the scattering angle between two states, $\phi = \angle(\mathbf{k}, \mathbf{p})$, is then “small” when both lie in the same flat patch, $|\phi| \lesssim \phi_0$.) For the screened Coulomb problem $R_s \sim 1/\kappa$, i.e. $\phi_0 \sim \sqrt{\kappa/k_F} = \sqrt{N\alpha}$; note that this is parametrically *larger* than $\phi_K = aN\alpha$ at weak coupling—we come back to this in 3.5.

2.2 The bosonized pair susceptibility. For $\phi < \phi_0$ the Fermi surface is divided into flat patches n with velocities $\mathbf{v}_n$, the density on each patch is a free boson, and a small-angle density–density interaction $V(q, i\nu)$ is quadratic in the bosons: the sector is Gaussian and exactly solvable (multidimensional bosonization). A fermion on patch n is $\psi_n \propto e^{i\Phi_n}$, and any correlator of fermion bilinears is a free-fermion correlator times a Debye–Waller factor. For the pair operator on opposite patches, $P = \psi_n \psi_{\bar{n}}$ ($\mathbf{v}_{\bar{n}} = -\mathbf{v}_n$), the phases add, $P \propto e^{i(\Phi_n + \Phi_{\bar{n}})}$, so

$$\chi_P(x, \tau) = \chi_P^{(0)}(x, \tau) e^{-Q_P(x, \tau)}, \quad Q_P = \frac{1}{2} \langle (\Phi_n + \Phi_{\bar{n}})^2 \rangle_{\text{conn}} = Q_{nn} + Q_{\bar{n}\bar{n}} + 2Q_{n\bar{n}}, \quad (11)$$

with the standard patch propagators

$$Q_{nm}(x, \tau) = \int \frac{d^2 q}{(2\pi)^2} \frac{d\nu}{2\pi} \frac{V_{\text{RPA}}(q, i\nu) [1 - \cos(\mathbf{q} \cdot \mathbf{x} - \nu\tau)]}{(i\nu - \mathbf{v}_n \cdot \mathbf{q})(i\nu - \mathbf{v}_m \cdot \mathbf{q})}. \quad (12)$$

(The particle–hole bilinear on a single patch, $\psi_n^\dagger \psi_n \propto e^{i(\Phi_n - \Phi_n)}$, has no Debye–Waller factor at all: the density is not renormalized by forward scattering, the Ward identity.) Whether the pair susceptibility is “dressed” is therefore the question whether $Q_P(x, \tau)$ grows without bound at large x, τ (a logarithm gives a power law, $\chi_P \propto \chi_P^{(0)}(r/R_s)^{-\gamma}$) or saturates (a finite factor, χ_P equal to the bare bubble up to a constant).

2.3 The plasmon window gives a finite Q_P . In the window $|\nu| > cv_F q$ both denominators in (12) are $\simeq (i\nu)^2 = -\nu^2$ for $m = n$ and for $m = \bar{n}$ alike, so the four terms of (11) add up, and the large-distance limit (where the cosine averages to zero) is, up to the sign convention of (12),

$$\begin{aligned} |Q_P^{(\text{pl})}| &\simeq 4 \int \frac{d^2 q}{(2\pi)^2} \int_{|\nu| > cv_F q} \frac{d\nu}{2\pi} \frac{2\pi\alpha}{q} \frac{\nu^2}{\nu^2 + \Omega_q^2} \frac{1}{\nu^2} = 4 \int_0^K \frac{q dq}{2\pi} \frac{2\pi\alpha}{q} \frac{1}{\pi\Omega_q} \arctan \frac{\Omega_q}{cv_F q} \\ &= \frac{4\alpha}{\pi} \int_0^K \frac{dq}{\Omega_q} \arctan \frac{1}{c\sqrt{2}q} \leq 2\alpha \int_0^K \frac{dq}{\Omega_q} = 4\sqrt{2}\alpha\sqrt{a}. \end{aligned} \quad (13)$$

The θ -function removes $|\nu| < cv_F q$ from the frequency integral, which replaces $1/2\Omega_q$ by $\frac{1}{\pi\Omega_q} \arctan(\Omega_q/cv_F q)$; numerically $|Q_P| = 2.4\alpha$ (1.9α) for $a = 0.3$, $c = 1$ ($c = 2$), compared with the bound $4\sqrt{2}\alpha\sqrt{a} = 3.1\alpha$. Either way the q -integral converges at small q because the pair weight $1/\nu^2$ is combined with the plasmon propagator $1/(\nu^2 + \Omega_q^2)$, whose frequency integral is $\propto q^{-1/2}$ —the same $\int d\xi/\sqrt{\xi}$ convergence found in the first note for the pairing kernel. There is no logarithm and no power law: within the plasmon window the small-angle sector multiplies the pair bubble by the constant $e^{-Q_P} = 1 + O(\alpha\sqrt{a})$. [The static part of the small-angle sector, $|\nu| < v_F q$, $V \rightarrow 2\pi\alpha/(q + \kappa)$, should be inserted here from the earlier bosonization note; for a static finite-range repulsion the reference result is a power-law suppression of χ_P with exponent $\gamma \propto V_2/v_F$ (Miserev, Schoeller, Klinovaja, Loss, arXiv:2312.17208, Eqs. 11, 29–30), i.e. a non-singular *suppression*, again nothing that could seed an instability.]

2.4 Consequence for the gap equation. The exact treatment of the small-angle sector says: the pair susceptibility dressed by $\phi < \phi_0$ scattering is the bare particle–particle bubble times a finite constant. The Cooper ladder over the same sector—which is what (7) is—instead generates the $\ln(\Omega_K/E)$ of Section 1.4 and with it $T_c \sim \exp(-c^2/\alpha^2)$. That logarithm is an artifact of applying the ladder where it does not apply. The correct procedure is therefore: subtract the small-angle sector from the gap equation (it has already been accounted for exactly, and it contributes nothing singular), and keep the Cooper ladder only for $\phi > \phi_0$, where it is equivalent to the one-loop RG. This is the gap equation of Section 3.

3. Gap equation with the small-angle sector removed

3.1 Kinematics of an angular cutoff. We integrate over $|\phi| > \phi_0$ in (2). The cut must be imposed in the variables in which it is defined, (ξ, ϕ) , and only then transformed. The integration region is

$$\mathcal{R} = \{(\xi, \phi) : |\phi| > \phi_0, \ q(\xi, \phi) < q_{\max}\}, \quad q^2 = \xi^2 + k_F^2 \phi^2.$$

Step 1 (fixed $\xi, \phi \rightarrow q$): $|\phi| > \phi_0$ means $q^2 > \xi^2 + k_F^2 \phi_0^2$, a ξ -dependent lower limit $q > \sqrt{\xi^2 + q_f^2}$, not $q > q_f$. Step 2 (exchange the order): $\{\sqrt{\xi^2 + q_f^2} < q < q_{\max}\}$ is the same set as $\{q_f < q < q_{\max}, |\xi| < b(q)\}$ with $b(q) \equiv \sqrt{q^2 - q_f^2}$, where

$$q_f \equiv k_F \phi_0 = \frac{\phi_0}{N\alpha} \quad (\text{units of } \kappa), \quad (14)$$

Thus transfers $q < q_f$ drop out entirely, *and* for $q > q_f$ the strips $b(q) < |\xi| < q$ drop out as well: these are the near-forward events at large $|\xi|$ that a plain lower cut $q > q_f$ would keep. The ξ -integral of (6) with the reduced range is again elementary ($\xi = q \sin \theta$, $\sin \theta_b = b/q$, $\tan \theta_b = b/q_f$):

$$\int_{-b}^b \frac{d\xi}{(w'^2 + \xi^2)\sqrt{q^2 - \xi^2}} = 2 \int_0^{\theta_b} \frac{d\theta}{w'^2 + q^2 \sin^2 \theta} = \frac{2}{|w'|\sqrt{w'^2 + q^2}} \arctan\left(\frac{\sqrt{w'^2 + q^2}}{|w'|} \frac{b}{q_f}\right), \quad (15)$$

which reduces to $\pi/|w'|\sqrt{w'^2 + q^2}$ for $q_f \rightarrow 0$. The factor $\frac{2}{\pi} \arctan(\cdot)$ is the fraction of the ξ -weight at fixed q that survives the angular cut. Its two limits show where the angle cut and a q -cut differ:

- $|w'| \ll q$ (Cooper regime): the ξ -integral is dominated by $|\xi| \lesssim |w'| \ll q$, i.e. by on-shell events with $|\phi| \simeq q/k_F$; the fraction $\rightarrow 1$ for $q > q_f$, and the angle cut and the q -cut coincide. The Cooper logarithm of events with $|\phi| > \phi_0$ is therefore fully retained—as it should be, since that is large-angle scattering.
- $|w'| \gg q$ (plasmon regime): the weight $1/\sqrt{q^2 - \xi^2}$ spreads ξ over the whole interval $(-q, q)$ (uniformly in $\theta = \arcsin(\xi/q)$), so every angle from 0 to q/k_F occurs, and the surviving fraction is $\frac{2}{\pi} \arccos(q_f/q) < 1$. Here the angle cut removes more than the q -cut.

Check. I compared (15) inserted into (6) with a direct two-dimensional integration over (ξ, ϕ) with the cut $|\phi| > \phi_0$ imposed literally, both with the small-angle kinematics (5) and with the exact ones, $q^2 = \xi^2 + 2k_F(k_F + \xi)(1 - \cos \phi)$ and measure $(k_F + \xi) d\xi d\phi$ ($k_F = 5\kappa$):

w'	ν	ϕ_0	$q_{\max}$	direct, exact	direct, small-angle	Eq. (15)	plain $q > q_f$
0.3	0.8	0.02	0.3	1.02463	1.02454	1.02454	1.605
0.2	0.9	0.05	0.3	0.24429	0.24368	0.24368	0.629
0.02	0.3	0.03	0.25	10.7335	10.7306	10.7306	12.24
0.01	0.5	0.02	0.3	76.5619	76.5479	76.5479	80.92
0.001	0.5	0.04	0.3	270.434	270.303	270.303	271.93

With the small-angle kinematics (5) the reduced formula is *identical* to the literal angular cut (agreement to 10^{-15} , i.e. Eq. (15) is exact, not an approximation); with the exact kinematics the difference is 10^{-4} – 3×10^{-3} , the curvature correction of relative order q/k_F , here ≤ 0.06 . (The direct integrals are done with the ϕ -limit where q reaches $q_{\max}$ computed at each ξ , so the integrand is smooth; with the exact kinematics this limit is ξ -dependent.) A plain lower cut on q is wrong by up to a factor 2.6 in the plasmon regime and becomes accurate only in the Cooper regime, exactly as the two limits above predict. In the gap equation the difference is quantitative, not qualitative: at $a = 0.3$, $E = 10^{-6} \kappa v_F$, $q_f = 0.1K$ ($0.3K$) the angle cut gives $\alpha = 0.97$ (1.84), the q -cut 0.87 (1.47). All results of this note use the angle cut (15). The gap equation becomes

$$\Delta(w) = -\frac{\alpha}{2\pi} \int dw' \int_{q_f}^{q_{\max}(\nu)} dq \frac{\nu^2}{\nu^2 + \Omega_q^2} \frac{2}{\pi} \arctan\left(\frac{\sqrt{w'^2 + q^2} \sqrt{q^2 - q_f^2}}{|w'| q_f}\right) \frac{\Delta(w')}{|w'| \sqrt{w'^2 + q^2}}, \quad (16)$$

with $q_{\max}(\nu) = \min(K, |\nu|/v_F)$: the upper limit enforces the plasmon condition $|\nu| > v_F q$ on the transfer (in addition to $q < K$), and the frequencies are restricted to $|w|, |w'| < 10 \Omega_K$. Note what (14) does: the cutoff is fixed in angle but, measured in units of the screening momentum, it moves with the coupling. The equation therefore depends on α both through its prefactor and through q_f .

3.2 An exact statement: the window must be open. Equation (16) has no solution unless $q_f < K$, i.e.

$$\alpha > \alpha_{\min}(\phi_0) = \frac{\phi_0}{Na} \quad (17)$$

For smaller α the screening momentum $\kappa = N\alpha k_F$ is so small that every plasmon transfer $q < a\kappa$ is a scattering angle below ϕ_0 : the whole plasmon window lies inside the exactly-solved sector, and there is simply nothing left for the ladder to act on. This is the origin of a critical coupling: it does not come from a competition of numbers in the kernel but from the kinematic fact that small-angle scattering, removed, takes the entire weak-coupling plasmon window with it.

3.3 Solution. For $q_f < K$, (16) is solved as before (logarithmic grid in w' , Gauss–Legendre in q , lowest eigenvalue μ of the matrix A defined by $\Delta = -\alpha A \Delta$; $\alpha = -1/\mu$). Since A depends on α only through q_f , it is convenient to compute $\mu(q_f)$ once on a grid of q_f/K and read off the curve parametrically: $\alpha_c = -1/\mu(q_f)$ and, from (14), $\phi_0 = N\alpha_c q_f$. Results for $N = 2$, $a = 0.3$ and 1, and infrared scales $E = 10^{-4}, 10^{-6}, 10^{-8} \kappa v_F$ are shown in Fig. 1a and in the table (table entries from solving $\alpha \mu(\phi_0/N\alpha) = -1$ directly at fixed ϕ_0).

α_c ($N = 2$)	$\phi_0 = 0.05$	0.1	0.2	0.3
$a = 0.3, E = 10^{-4}$	1.19	1.42	1.78	2.08
$a = 0.3, E = 10^{-6}$	0.93	1.15	1.48	1.76
$a = 0.3, E = 10^{-8}$	0.81	1.01	1.32	1.59
$a = 1.0, E = 10^{-4}$	0.77	0.87	1.03	1.16
$a = 1.0, E = 10^{-6}$	0.60	0.70	0.84	0.96
$a = 1.0, E = 10^{-8}$	0.51	0.61	0.74	0.86
bound (17), $a = 0.3$	0.08	0.17	0.33	0.50
bound (17), $a = 1.0$	0.025	0.05	0.10	0.15

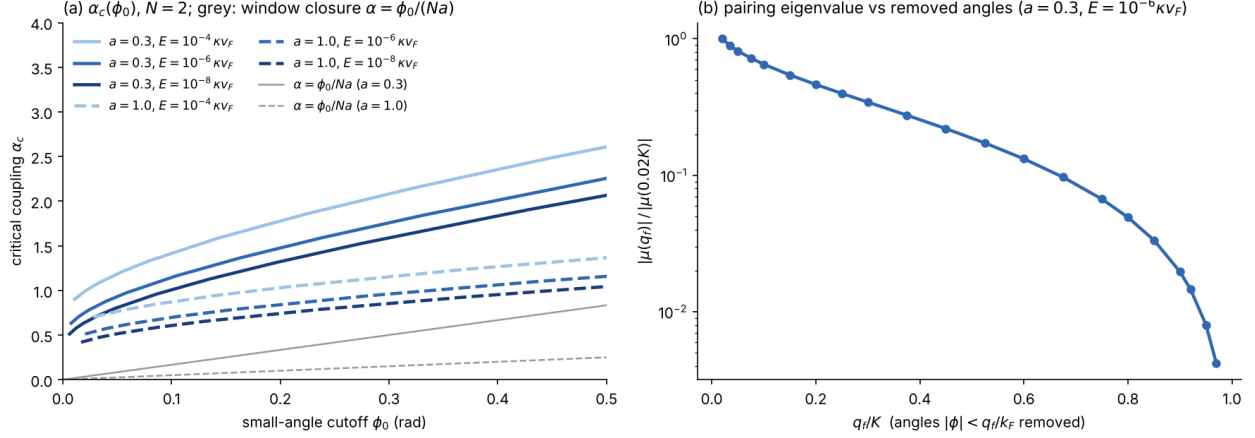


Figure 1: (a) Critical coupling of the gap equation (16) as a function of the small-angle cutoff ϕ_0 , for $N = 2$, two window sizes a and three infrared scales E , plotted parametrically, $\alpha_c = -1/\mu(q_f)$, $\phi_0 = N\alpha_c q_f$. Grey lines: the exact bound (17). (b) The pairing eigenvalue of the naive equation (7) when the angles $|\phi| < q_f/k_F$ are removed (angular cut, Eq. (15)), relative to its value at $q_f = 0.02K$ ($a = 0.3, E = 10^{-6} \kappa v_F$): the solution of the naive equation is built from the smallest angles.

3.4 Reading the result. Three features. (i) α_c is of order one for any reasonable ϕ_0 : at $\phi_0 = 0.1$ it is 0.6–1.4 depending on a and E . There is no plasmon-mediated superconductivity at weak coupling. (ii) α_c increases with ϕ_0 roughly like $\phi_0^{1/3}$ over the tabulated range (local exponent 0.23–0.36 between $\phi_0 = 0.05$ and 0.3, larger for smaller a and E), and decreases with the size of the window roughly like $a^{-1/2}$ (between $a^{-0.4}$ and $a^{-0.5}$) (from $\Omega_K \propto \sqrt{a}$, as in the pure-window estimate of the earlier note). (iii) At fixed ϕ_0 , α_c still decreases slowly with the infrared scale E . This is the second-order Tolmachev–Anderson mechanism of Section 1.4 operating inside the *surviving* part of the window, $q_f < q < K$ (angles between ϕ_0 and ϕ_K): for those transfers the Cooper ladder is legitimate, so the $\ln(\Omega_K/E)$ is real, and $\alpha_c(E)$ approaches the bound (17) as $E \rightarrow 0$, but only like $[\ln(\Omega_K/E)]^{-1/2}$, i.e. never in practice: at $\phi_0 = 0.1, a = 0.3$ the bound is 0.17, while $\alpha_c = 1.0$ at $E = 10^{-8} \kappa v_F$. The practical statement is therefore $\alpha_c \simeq 1$, with the exact bound (17) as the $T = 0$ limit.

3.5 Closing the loop: ϕ_0 for the screened Coulomb problem. With $R_s \sim 1/\kappa$ in (10), $\phi_0 \sim \sqrt{\kappa/k_F} = \sqrt{N\alpha}$, and the bound (17) becomes a condition on α alone: $\alpha > \sqrt{N\alpha}/Na \Leftrightarrow N\alpha > 1/a^2$, i.e.

$$\alpha_c \gtrsim \frac{1}{Na^2} \quad (R_s \sim 1/\kappa), \quad (18)$$

which is 0.5 for $N = 2, a = 1$ and 5.6 for $a = 0.3$ —order one at best, and this is only the $T = 0$ bound; the finite- E values of Fig. 1a lie above it. Equivalently: at weak coupling $\phi_K = aN\alpha \ll \sqrt{N\alpha} = \phi_0$, so the *entire* plasmon window is inside the bosonizable sector. The identification $R_s \sim 1/\kappa$ is the natural one for the statically screened interaction; for the dynamically screened one the range at transfer frequency ν is $1/q_\nu$, which is longer still and only strengthens the conclusion.

4. Conclusions

1. The BCS-like gap equation for the plasmon interaction, Eq. (7), is built entirely from scattering angles $\phi \leq aN\alpha \ll 1$ (Eqs. 4, 9, Fig. 1b). Its solution at arbitrarily weak coupling, $T_c \sim \Omega_K e^{-c^2/\alpha^2}$, is a Cooper-ladder logarithm generated by small-angle scattering.

2. Small-angle scattering, $\phi < \phi_0 = (k_F R_s)^{-1/2}$, is exactly solvable by bosonization, and the exact result is that it does not dress the pair susceptibility in any relevant way: in the plasmon window the Debye–Waller exponent of the pair correlator is the finite number (13), $O(\alpha\sqrt{a})$, with no logarithm. The Cooper ladder is the wrong resummation for this sector and must be replaced by the exact (trivial) answer; the ladder is kept only for $\phi > \phi_0$, where it is the one-loop RG.
3. The resulting gap equation (16) has no solution below $\alpha_{\min} = \phi_0/Na$ (the window closes), and in practice a critical coupling $\alpha_c \simeq 1$ (Fig. 1a). With $R_s \sim 1/\kappa$ the bound alone gives $\alpha_c \gtrsim 1/Na^2$. Plasmon-mediated superconductivity at weak coupling is excluded; at $\alpha \gtrsim 1$ the window reaches large angles and the one-loop treatment is no longer controlled.